# SciCLIP: Parameter-Efficient Fine-Tuning of CLIP for Cross-Modal Scientific Figure Retrieval

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## Abstract

Retrieving scientific figures (plots, diagrams, and charts) from natural language queries is a practical but difficult cross-modal retrieval problem. General-domain vision-language models such as CLIP perform well on natural images but exhibit a considerable domain gap when applied to academic figures with specialized visual and textual semantics. In this paper, we describe **SciCLIP**, a lightweight adaptation method that applies Low-Rank Adaptation (LoRA) to a frozen `clip-vit-base-patch32` backbone, targeting the query, key, and value projection layers of both encoders.

We train on an 18,000-sample subset of the SciCap dataset and evaluate on a 2,000-candidate retrieval pool. The adapted model improves Median Rank from 110.0 to 34.0 and MRR from 0.1564 to 0.2324 relative to the zero-shot baseline. While absolute retrieval accuracy remains moderate, these results indicate that even minimal adaptation ($\sim$0.1% trainable parameters) can meaningfully reduce the domain gap. We additionally present an ablation over LoRA ranks ($r \in \{4, 8, 16\}$), observing that lower ranks provide a useful regularizing effect under limited training data, and describe a Gradio-based interactive retrieval demo backed by a FAISS index.

## 1 Introduction

Cross-modal information retrieval—searching for visual content using natural language queries—has advanced considerably through joint vision-language representation learning. In academic contexts, scientific figures often summarize key results of research papers, and the ability to retrieve them via textual queries has applications in literature review, survey generation, and academic search.

However, scientific figure retrieval poses distinct challenges. Models like CLIP (Radford et al., 2021), trained on large-scale natural image–text pairs, lack exposure to the specialized vocabulary and visual conventions of academic diagrams. Terms such as "LDPC code bit error ratio" or "t-SNE embedding scatter plot" have little representation in general web corpora, and the visual structure of scientific figures (axes, legends, schematic boxes) differs substantially from natural photographs.

Full fine-tuning of large vision-language models on domain-specific data is resource-intensive and risks catastrophic forgetting. Parameter-Efficient Fine-Tuning (PEFT) methods offer an alternative. In particular, Low-Rank Adaptation (LoRA) (Hu et al., 2022) injects small trainable matrices into frozen transformer layers, allowing domain adaptation with minimal parameter overhead.

In this work, we apply LoRA-based adaptation to CLIP for scientific figure retrieval and report the following contributions:

- **Data Pipeline:** We describe a preprocessing pipeline for the `CrowdAILab/scicap` dataset that addresses practical challenges including multi-volume ZIP extraction bottlenecks. By sorting extraction targets by physical byte offset, we reduce preprocessing time from over 20 minutes to under 2 minutes.
- **LoRA-Adapted Retrieval:** We show that adapting $\sim$0.1% of CLIP's parameters with LoRA produces consistent improvements over the zero-shot baseline across all measured retrieval metrics.
- **Rank Ablation:** We compare LoRA ranks $r \in \{4, 8, 16\}$ and find that lower ranks tend to generalize better under our limited training regime, while higher ranks show signs of diminishing returns.
- **Interactive Demo:** We provide a Gradio-based side-by-side comparison interface using FAISS for efficient nearest-neighbor search.

## 2 Related Work

### 2.1 Vision-Language Pretraining and CLIP

CLIP (Radford et al., 2021) learns joint vision-language representations through contrastive learning on a large corpus of image–text pairs. Its twin-tower architecture projects images and text into a shared embedding space, enabling zero-shot transfer to a range of downstream tasks. However, because its training data consists predominantly of natural images with general web text, performance degrades on domain-specific content such as scientific figures.

### 2.2 Parameter-Efficient Fine-Tuning and LoRA

LoRA (Hu et al., 2022) adapts pre-trained models by decomposing weight updates into low-rank matrices. For a frozen weight matrix $W_0 \in R^{d\times k}$, the adapted weight is:

$$W = W_0 + \frac{\alpha}{r} B \cdot A \tag{1}$$

where $B \in R^{d\times r}$, $A \in R^{r\times k}$, and $r \ll \min(d, k)$. This formulation has been applied to language models and diffusion models; its use for vision-language retrieval in specialized domains remains less explored.

### 2.3 Scientific Figure Datasets

The SciCap dataset (Hsu et al., 2021) pairs scientific figures with their captions. Working with SciCap presents practical difficulties: the dataset is distributed as multi-volume ZIP archives with column mismatches across splits. We address these issues through careful preprocessing, as described in Section 3.2.

## 3 Methodology

### 3.1 Model Architecture and Adapter Placement

We use `openai/clip-vit-base-patch32` as the backbone. The visual encoder (ViT-B/32) maps $224 \times 224$ images to 512-dimensional vectors; the text encoder maps token sequences (max length 77) to vectors of the same dimensionality.

LoRA adapters are injected into the query, key, and value projection layers (`q_proj`, `k_proj`, `v_proj`) of both the visual and textual attention blocks. All original weights remain frozen. At rank $r = 8$, the trainable parameters constitute approximately 0.1% of the total model parameters.

### 3.2 Dataset Pipeline and Preprocessing

We preprocess the `CrowdAILab/scicap` dataset as follows. Annotations are loaded from `train.json` in COCO format, and image IDs are mapped to file names.

- **Caption Index Removal:** Captions frequently begin with structural prefixes (e.g., "Figure 3:" or "Fig. 4(b):") that can serve as superficial shortcuts during training. We use the `caption_no_index` field to strip these prefixes, encouraging the model to attend to semantic content.

- **Length Filtering:** We discard captions shorter than 20 characters or longer than 300 characters. Short captions lack informative content; long captions exceed CLIP's 77-token input limit.

- **Sequential Extraction:** The SciCap images are distributed across 11 split ZIP volumes totaling approximately 20 GB. Naive random-access extraction incurs severe seek overhead on cloud storage. We merge the volumes using `zip -F` and sort extraction targets by their `header_offset`, processing files in physical order. This reduces extraction time from over 20 minutes to under 2 minutes.

### 3.3 Training Objective

We train with the symmetric InfoNCE contrastive loss(van den Oord et al., 2018). For a batch of $N$ image–text pairs with normalized embeddings $\{v_i\}_{i=1}^N$ and $\{t_i\}_{i=1}^N$, the pairwise similarity is:

$$S_{i,j} = e^{\tau \cdot (v_i^\top t_j)} \tag{2}$$

where $\tau$ is a learned temperature. The loss averages the image-to-text and text-to-image directions:

$$\mathcal{L}_{v2t} = -\frac{1}{N} \sum_{i=1}^{N} \log \frac{S_{i,i}}{\sum_{j=1}^{N} S_{i,j}} \tag{3}$$

$$\mathcal{L}_{t2v} = -\frac{1}{N} \sum_{i=1}^{N} \log \frac{S_{i,i}}{\sum_{j=1}^{N} S_{j,i}} \tag{4}$$

$$\mathcal{L}_{total} = \frac{\mathcal{L}_{v2t} + \mathcal{L}_{t2v}}{2} \tag{5}$$

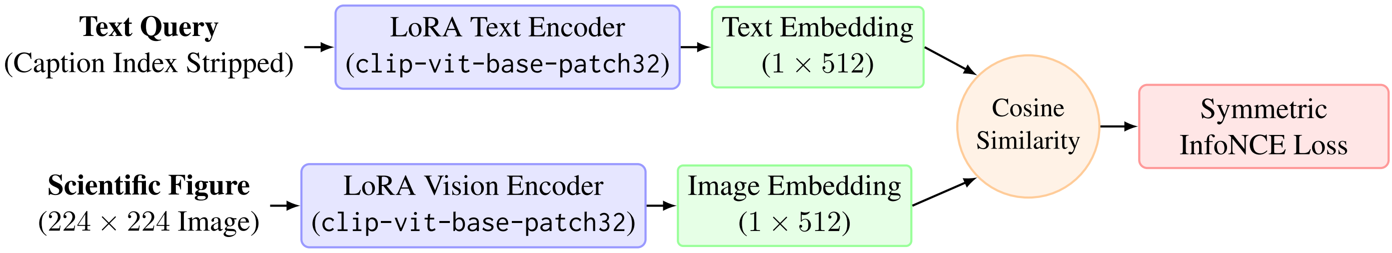


Figure 1: SciCLIP training pipeline. LoRA adapters are applied to the query, key, and value projection layers of both encoders. The model is trained with a symmetric InfoNCE loss over paired scientific figures and captions.

## 4 Experiments

### 4.1 Setup

Training is conducted on an NVIDIA A100 GPU via Google Colab. From 20,000 preprocessed samples, we use 18,000 for training and 2,000 for test (90/10 split). We train for 5 epochs with batch size 64, using the AdamW optimizer (learning rate $5 \times 10^{-5}$, weight decay 0.2) with cosine annealing and 50 warmup steps.

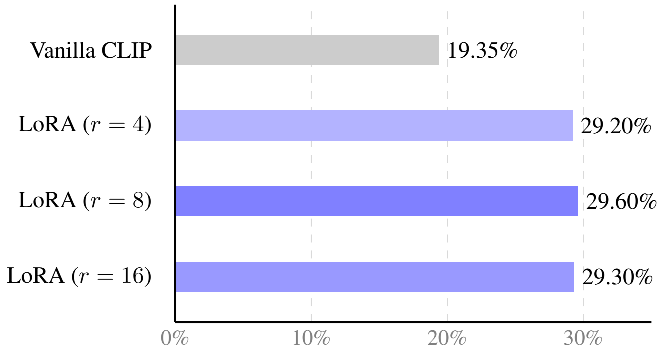


Figure 2: Recall@5 by model configuration. LoRA adaptation consistently improves over the baseline; differences among rank settings are small.

### 4.2 Metrics

We evaluate using standard retrieval metrics on the 2,000-candidate pool:

- **Recall@k:** Fraction of queries for which the correct image appears in the top $k$ results ($k \in \{1, 5, 10\}$).
- **Mean Reciprocal Rank (MRR):**

$$\mathrm{MRR} = \frac{1}{M} \sum_{i=1}^{M} \frac{1}{\mathrm{rank}_i}$$

- **Median Rank:** Median position of the correct image across all queries (lower is better).

### 4.3 Results

Table 1 summarizes the retrieval performance.

### 4.4 Analysis

We observe the following patterns:

1. **Baseline improvement:** All three LoRA configurations improve over the zero-shot baseline. Recall@5 increases from 19.35% to roughly 29–30%, and the Median Rank drops from 110 to 34–36. While these relative gains are notable, the absolute recall values (e.g., R@1 of 16–17%) indicate that the task remains challenging and the adaptation alone does not fully close the domain gap.
2. **Effect of rank on generalization:** The smallest rank ($r = 4$) achieves the highest R@1 (0.1675) and MRR (0.2328), while $r = 8$ yields the best R@5 (0.2960). These differences are small, but the pattern is consistent with the hypothesis that lower-rank adapters act as an implicit regularizer when training data is limited. With only 18,000 samples, a more constrained adapter may be less prone to fitting noise in the training captions.
3. **Diminishing returns at higher rank:** The $r = 16$ configuration does not improve over $r = 8$ on any metric and shows slight degradation on R@1 and MRR. This suggests that under our training budget, increasing adapter capacity beyond $r = 8$ offers no benefit and may lead to mild overfitting.

## 5 Qualitative Examples and Demo Interface

To complement the quantitative evaluation, we inspect selected retrieval results.

- **Query:** *"attention mechanism transformer architecture diagram."* Vanilla CLIP returned general block diagrams and unrelated line

| Model | R@1 ↑ | R@5 ↑ | R@10 ↑ | MRR ↑ | Med. Rank ↓ |
|---|---|---|---|---|---|
| Vanilla CLIP (zero-shot) | 0.1090 | 0.1935 | 0.2380 | 0.1564 | 110.0 |
| SciCLIP ($r = 4$) | **0.1675** | 0.2920 | **0.3585** | **0.2328** | 36.0 |
| SciCLIP ($r = 8$) | 0.1655 | **0.2960** | 0.3545 | 0.2324 | **34.0** |
| SciCLIP ($r = 16$) | 0.1640 | 0.2930 | **0.3585** | 0.2312 | **34.0** |

Table 1: Cross-modal retrieval results on the 2,000-candidate test set. All LoRA-adapted variants improve over the zero-shot baseline, though absolute recall values remain moderate.

plots, likely matching on surface-level terms like "architecture." SciCLIP retrieved a transformer encoder-decoder diagram with multi-head attention blocks, suggesting improved alignment between technical vocabulary and visual structure.

- **Query:** *"training loss curve comparison baseline."* Vanilla CLIP returned bar charts and tables containing the word "comparison." SciCLIP returned line plots showing training and validation loss trajectories, a more semantically appropriate match.

These examples illustrate that the adapted model captures domain-specific associations that the zero-shot model misses, though we note that qualitative examples are inherently selective.

To enable interactive verification of these retrieval examples, we also deploy a Gradio-based side-by-side interface backed by a FAISS index, achieving sub-millisecond search latencies

## 6 Limitations

Several limitations should be noted. First, our training and test sets are drawn from the same 20,000-sample subset of SciCap; we do not evaluate on a separately held-out test set, which limits claims about generalization. Second, the absolute retrieval accuracy remains low—an R@1 below 17% means the correct figure is the top result less than one-fifth of the time. This reflects both the inherent difficulty of the task (many scientific figures are visually similar) and the limited scale of our training data. Third, we evaluate only a single backbone (`clip-vit-base-patch32`); larger or more recent vision-language models may yield different conclusions about the effectiveness of LoRA adaptation. Finally, our rank ablation covers only three values; a finer-grained sweep and multiple random seeds would strengthen the analysis.

## 7 Conclusion

We presented SciCLIP, a LoRA-based adaptation of CLIP for scientific figure retrieval. With approximately 0.1% trainable parameters, the method reduces the Median Rank of correct retrievals from 110 to 34 and improves MRR from 0.1564 to 0.2324 on a 2,000-candidate test set. An ablation over LoRA ranks suggests that smaller ranks ($r = 4, 8$) generalize better than $r = 16$ under limited training data, consistent with the view that low-rank constraints provide implicit regularization. These results represent a starting point rather than a complete solution. The moderate absolute performance highlights the difficulty of scientific figure retrieval and the need for larger-scale training data, stronger backbones, and possibly hybrid retrieval approaches. In future work, we plan to scale training to the full SciCap dataset (approximately 330,000 samples) and explore incorporating OCR-extracted text from within figures to capture fine-grained textual information that image encoders alone may miss.

## References

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## A Hyperparameter Details

Table 2 lists the hyperparameters used in our experiments.

| **Hyperparameter** | **Value** |
|---|---|
| Base Model | `openai/clip-vit-base-patch32` |
| Optimizer | AdamW |
| Learning Rate | $5 \times 10^{-5}$ |
| Weight Decay | 0.2 |
| Batch Size | 64 |
| Epochs | 5 |
| Warmup Steps | 50 |
| LR Scheduler | Cosine Annealing |
| LoRA Target Modules | `q_proj`, `k_proj`, `v_proj` |
| LoRA Alpha ($\alpha$) | 16 |
| LoRA Dropout | 0.1 |

Table 2: Training hyperparameters.